Integer Planar Cores for Deterministic Certificates of Grammatical Distribution Drift

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Abstract

Structured diffusion decoders often condition a frozen product prediction on a recursive grammar. Comparing two such predictions requires measuring their conditioned distributions, rather than comparing independent token marginals. We give an integer implementation of planar support compression with $O(\eta^{-1/2})$ generators and near-linear construction cost. Combined with known admissible-gate propagation, this yields deterministic rational relative total-variation certificates on unambiguous event forests, with polynomial bit complexity and no positive-mass or positive-distance floor. For ordinary products the arithmetic bound is $O(nq\sqrt{n/\varepsilon}\log(qn/\varepsilon+2))$, improving the precision exponent of the stated deterministic bounds examined here. A classical chart construction gives a full-input polynomial corollary for trimmed LL(1) grammars and original token identities. A complete six-case audit on one public diffusion-model head illustrates probability-table compression: the FP16 certificate is substantially tighter than inexpensive certified Hellinger and likelihood-range controls. The geometric and inference principles are antecedents; the contribution is quantitative and concerns distribution auditing, without claiming faster generation or semantic gains.

1 Problem and quantitative contribution

Let $S_1, \dots, S_n$ be finite domains of original token identities. Each token has a fixed byte emission. A canvas and an unambiguous grammar specify a common valid set $\mathcal{F} \subseteq \prod_i S_i$. For two nonnegative rational tables p_i, q_i with row sums at most one, let

$$Z_p = \sum_{y\in\mathcal{F}} \prod_i p_i(y_i), \qquad P(y) = \frac{\prod_i p_i(y_i)}{Z_p},$$

and define Z_q, Q analogously. We require $Z_p, Z_q > 0$; otherwise this comparison is undefined. Zeros, token aliases, fixed positions and rare valid events are allowed. The target is a rational interval

$$L \le \mathrm{TV}(P,Q) \le U \le (1+\varepsilon)L, \qquad 0 < \varepsilon \le 1. \tag{1}$$

This is the frozen, grammar-conditioned product law of a step, not the learned joint law of a complete neural trajectory. No EOS is admitted here.

TV as a support query, admissible-gate composition and deterministic relative approximation are established principles [1]. Classical convex approximation gives the planar exponent [3]. Our construction retains original integer polygon endpoints; its generators have no new denominators, and its nearest-edge queries use exact integer comparisons. These properties provide the explicit arithmetic and bit bounds below. We do not claim a new geometric approximation rate or a TV lower bound.

Feng, Liu and Liu's product FPTAS has stated arithmetic cost [2, Theorem 1]

$$O(qn^2\varepsilon^{-1}\log q \log(n/(\varepsilon\,\mathrm{TV}))).$$

Its $(1-\varepsilon)$ convention converts to (1) by using error $\varepsilon/(1+\varepsilon)$. Fu's paired product corollary gives $\widetilde{O}(nq(n/\varepsilon)^4)$ and already removes numerical TV dependence in an extended real-arithmetic model [1, Corollary 1.4]. Our sharper product bound uses precision exponent $1/2$ and polynomial-bit integers. Comparisons are of published upper bounds, not lower bounds on competent alternative implementations or guarantees of physical speed.

Randomized inference-based relative approximation is also an antecedent [4]; a derivation encoding supplies a relevant alternative, as discussed in Section 6. Randomization may be preferable when a probabilistic failure bound is acceptable. The finite numerical audit does not compare those algorithms' runtimes.

2 An exact integer planar core

For a finite multiset G of integer pairs, put

$$F_G(w) = \sum_{g \in G} |g \cdot w|, \qquad Z(G) = \sum_{g \in G} [-g, g].$$

Thus F_G is the support function of the centrally symmetric polygon $Z(G)$.

Theorem 1 (Integer endpoint compression)**.** *Given m integer pairs and integer $k \geq 1$, one can construct a multiset C of at most $64k$ integer pairs such that, for every $w \in \mathbb{R}^2$,*

$$(1 - (16k^2)^{-1}) F_G(w) \leq F_C(w) \leq F_G(w). \tag{2}$$

The construction uses $O((m+k)\log(m+2))$ exact arithmetic operations. Each output coordinate is a signed sum of input coordinates. Rank zero or one is compressed exactly. Intermediate integers have polynomial input bits.

Proof. Orient nonzero generators into a half-plane and sort by determinant. Starting at minus their sum and walking by twice each generator, then its opposite, enumerates $Z(G)$. Every polygon vertex has the same coordinate parities. Redundant collinear vertices may be removed. For rank two choose a vertex u of maximum squared Euclidean norm and a vertex v of maximum $|\det(u, v)|$, reversing v to make $D = \det(u, v) > 0$. If $T = [u\ v]^{-1}$ and $K = TZ(G)$, then

$$\operatorname{conv}(\pm e_1, \pm e_2) \subseteq K \subseteq [-2, 2] \times [-1, 1]. \tag{3}$$

Indeed the coefficient $\beta = \det(u, x)/D$ is at most one in absolute value. Writing $x = \alpha u + \beta v$, maximality of $\|u\|$ gives $|\alpha|\|u\| \leq \|x\| + \|v\| \leq 2\|u\|$. The inner inclusion follows from the four chosen vertices. Hence $h_K(w) \geq \|w\|_\infty$.

Take the $32k$ points of spacing $1/k$ on the boundary of $[-4, 4]^2$. For each, find a closest polygon edge and retain both of its endpoints and their opposites. Let R be their convex hull. Clearly $R \subseteq K$. For a unit supporting normal w, let x maximize $w \cdot x$ on K. The ray $x + tw$ reaches the outer square at p, with $t \geq 2$. A grid point q on that side has $\|q - p\| \leq 1/(2k)$. Let y be the projection of q onto K. Projection optimality gives

$$(q - y) \cdot (x - y) \leq 0, \qquad t\, w \cdot (x - y) \leq -\|x - y\|^2 - (q - p) \cdot (x - y) \leq \frac{\|q - p\|^2}{4}.$$

Because y belongs to a retained edge, it is in R. Thus $h_K(w) - h_R(w) \leq 1/(32k^2)$. Since $h_K(w) \geq 1/2$ for unit w, homogeneity gives (2) for these two polygons.

Map the retained vertices back by T^{-1}. Their centered hull has $2r \leq 128k$ vertices $z_0, \ldots, z_{2r-1}$ in boundary order, with $z_{i+r} = -z_i$. Its generators are $b_i = (z_{i+1} - z_i)/2$, $0 \leq i < r$. They are integers by the common parity, and each is the sum of a consecutive run of oriented original generators. Their starting vertex is $-\sum_i b_i = z_0$, so they represent the hull exactly. One may return the original list if it is shorter.

Polygon construction and retained-hull construction cost $O(m \log(m+2))$. Constant-factor normalization uses two linear scans. Lemma 1 handles each outside query in logarithmic time, giving the stated total. After scaling by kD, all geometry uses integer cofactors, determinants, squared segment distances and cross multiplication. No projection coordinate is output and no new rational denominator is introduced. □

Lemma 1 (Outside queries)**.** *For a convex polygon strictly to the left of $q = (4, Y)$, after scaling as above, an exact nearest edge is found in logarithmic arithmetic time with linear preprocessing. Quarter-turns handle the other three sides.*

Proof. The projection lies on the right boundary chain, from the rightmost lowest vertex to the rightmost highest vertex. Along this chain the second coordinate increases strictly after collinear removal. Write the boundary as $x = f(t)$, where f is concave and $4 - f(t) > 0$. Then

$$D(t) = (4 - f(t))^2 + (Y - t)^2$$

is strictly convex. Its vertex values decrease then increase even with nonuniform vertex spacing. Binary search finds a minimum vertex; one of its two neighboring segments contains the continuous minimizer. Compare their exact squared distances by integer cross multiplication. An edge tie still contains the unique projection. Four rotated chains need only linear space and preparation. □

The $O(\eta^{-1/2})$ order is classical worst-case optimal for approximating *all* planar support directions: an inscribed $2r$-vertex polygon has an angular gap at least π/r, and a circle loses order r^{-2} support at that gap's midpoint. Fine rational polygons approximate the circle. This is not a lower bound for computing one TV query.

3 Event forests, relative error and bit cost

Definition 1. *An admitted event forest is an acyclic sum/product circuit with a fixed set of original positions, or scope, at each node. A leaf lists distinct choices at one position, or is an empty-scope unit/zero. Product scopes are disjoint and their events form a Cartesian product. Sum alternatives have equal scopes and disjoint assignment events. The root represents each member of $\mathcal{F}$ once. Productions have neutral weights. Empty-scope nodes reduce to unit/zero. There is no common-v-tree requirement.*

For each head independently let D_i clear all denominators at position i, and let $a_i(v) = D_i p_i(v)$; use separate denominators and b_i for q. Compute the exact positive integer root masses A, B by ordinary inside addition/multiplication. They are not inferred from compressed generators. At a leaf keep pairs $(a_i(v), b_i(v))$; at a sum concatenate child lists; at a product take all componentwise products of child pairs. Compress each internal gate by Theorem 1. Core pairs may be signed: they represent norms, never probabilities or a token sampler. The rounded head Q remains the actual alternative law.

Let d be the maximum number of compression gates in an expanded root derivation: a sum takes the maximum child count plus one; a binary product adds both child counts plus one. Use $d \geq 1$. Set $k = \lceil \sqrt{d/\varepsilon} \rceil$, computed by integer square root and comparisons. At the root return

$$L = \frac{\sum_{(x,y)\in C} |Bx - Ay|}{2AB}, \qquad U = \min\{1, (1+\varepsilon)L\}. \tag{4}$$

Theorem 2 (Deterministic rational forest certificate)**.** *The algorithm satisfies* (1)*, including TV zero exactly. Let H count explicit nodes and wires. Then $d \leq H$. With bounded fan-in and bounded exact leaf lists, its core arithmetic cost and storage are*

$$O(H(d/\varepsilon)\log(d/\varepsilon + 2)), \qquad O(H\sqrt{d/\varepsilon} + d/\varepsilon).$$

For general leaf lists of maximum size V, set $M = 64k$, $K_0 = \max(V, M)$ and $J = HK_0 + K_0^2$. A conservative full bit bound is

$$\mathrm{poly}(\ell) + O\big(H(J+k)\log(J+2)(\ell + \log(J+k+2))^2\big),$$

where ℓ is total input bit length. No conditioning-mass or TV floor is needed. Preparation of an input representation is charged separately.

Proof. If two child cores preserve all query norms from below by factors α and β, disjoint sum preserves their smaller factor. For a product, fix each right generator and query the left norm with its componentwise product with w. Sum the inequalities, then do the same for the right norm using the retained left generators. This gives lower factor $\alpha\beta$ and upper factor one, even with signed retained pairs. Induction gives $(1 - 1/(16k^2))^d F_{\mathrm{exact}} \le F_C \le F_{\mathrm{exact}}$. Bernoulli's inequality and $k^2 \ge d/\varepsilon$ imply a lower factor at least $1 - \varepsilon/16 \ge 1/(1+\varepsilon)$. At query $w = (1/A, -1/B)$ the exact norm is $2\,\mathrm{TV}(P, Q)$, proving the interval.

A nonempty scoped node cannot repeat in a derivation: its lowest common ancestor would be a product whose two disjoint scopes contain that same nonempty scope. Acyclicity prevents repetition along a chain. This proves $d \le H$; repeated-squaring arithmetic DAGs are excluded by the scope contract.

Distinct-event mass at scope I is at most $\prod_{i \in I} D_i$, since each input row sums to at most one. Evaluate core containment on either coordinate axis: the sum of absolute coordinates is at most that exact head mass. This invariant also bounds product/union intermediates before compression. Each coordinate therefore has $O(\ell)$ bits. Polygon vertices, cofactors, scaled grid points and squared-distance comparisons have $O(\ell + \log(J+k+2))$ bits, up to constant factors. Standard integer multiplication gives the displayed conservative bit bound. Initial LCMs, normalizers and final rational operations contribute polynomial input work. At most J candidates occur at a gate; Theorem 1 then bounds every compression. Under bounded arities and leaf sizes products have $O(M^2)$ rows and sums $O(M)$, giving the sharper expressions. □

Corollary 1 (Products)**.** *For two rational product measures on n domains of size at most q, the arithmetic cost is*

$$O\big(nq\sqrt{n/\varepsilon}\log(qn/\varepsilon + 2)\big)$$

plus input/scaling/normalizer work, with polynomial-bit arithmetic.

Proof. Start with $(1, 1)$. At position i, concatenate the diagonal images $(a_i(v)x, b_i(v)y)$ for every original v and every previous core pair (x, y). Compress once. Here $d \le n$ and the intermediate list has qM pairs; it does not form a wasteful general M^2 Cartesian product. Normalizer masses are products of integer row sums. Apply the preceding norm induction and nearest-edge bound. For an empty product TV is zero. □

4 A polynomial original-word grammar corollary

The forest-input theorem is not a proof of a software normalizer. Here is a separate mathematical full-input construction. Trim unreachable and unproductive symbols of an LL(1) byte grammar. Keep unit/epsilon rules, replace terminals in longer bodies by preterminals, and right-binarize with fresh fixed-chain auxiliaries. This is linear in grammar symbol occurrences g and preserves derivations bijectively, without nullable-subset expansion.

Construct an acyclic byte-prefix trie per original token position with a separate final-byte closure edge for each original ID. Prefix tokens close to the next boundary while longer tokens continue within the current trie. Aliases have distinct closure edges; their probabilities occur once on those edges. Empty/unsupported emissions are invalid and keep their mass in the input table. Complete lattice paths and original assignments are bijective. Let V_b, E_b be lattice vertices/edges; these are bounded by total declared token-byte occurrences before sharing.

Theorem 3 (LL(1) source-input certificate)**.** *For the trimmed LL(1) source, original token domains and positive valid masses above, a deterministic rational certificate* (1) *is computed with total cost polynomial in* $g, V_b, E_b, \ell, 1/\varepsilon$*. No common parse tree or fixed maximum nesting depth is assumed.*

Proof. For a binary rule $A \to BC$ add same-span dependency $A \to B$ if C is nullable and $A \to C$ if B is nullable; add unit dependencies as well. Every dependency is a nullable-prefix left corner. A cycle collapses to source left recursion, excluded by LL(1); auxiliary chains alone cannot cycle.

Process chart cells (A, u, v) by increasing topological difference $v - u$ and then in dependency order. At $u = v$ use the unique empty derivation or zero. At positive spans include matching byte edges, unit children and binary splits at every $u \le w \le v$. Interior children have shorter spans; endpoint children use a nullable unit and an already computed same-span child. Hence the circuit is acyclic and has $O(gV_b^2)$ cells and $O(gV_b^3 + gE_b)$ alternatives.

Standard chart induction gives every path parse exactly once. LL(1) unambiguity makes alternatives disjoint as original assignment events, not just tree tags. Trimming ensures relevant nonterminal ambiguity cannot be hidden outside the start language. Vertex layers determine consumed closure slots, so sum scopes agree and product scopes partition. An empty scope stays inside one trie, with a unique path and unambiguous parse, and reduces to unit/zero. Theorem 2 applies. All source/lattice work is charged. This is classical parsing used for admission, not a new parsing algorithm. □

5 A software use: auditing numerical probability tables

If a certified upper bound U is below an application's error budget τ, replacing P by Q changes the expectation of *every* verifier with values in $[0, 1]$ by at most τ. This is the standard operational meaning of TV; it requires no chosen semantic labels or favorable benchmark. For numerical table storage, nominal FP64-to-FP32 storage halves and FP64-to-16-bit storage quarters before metadata and transfers. We certify the distributional effect, not model throughput or parameter quantization.

A common continuation kernel cannot increase TV. For a finite iterative decoder with uniform per-reachable-history certificates τ_t, a coupling gives final trajectory distance at most $\sum_t \tau_t$. This consequence requires the same sampling/commitment policy. A confidence threshold that changes under rounding needs a separate policy-preservation argument; local head closeness alone does not prove trajectory closeness. One observed history also does not certify every possible future history.

5.1 Complete functional audit, not a competitive corpus

The public packet contains an archived full MDLM head on a 19-slot canvas, four free positions and all 50,258 original IDs. Its selection preceded the independent A24 observations; it is the sole complete packet published offline. No new model forward, changed source completion or semantic label is used. We froze all three rounding representations (FP32, FP16, BF16) and both tolerances before measurement. Round *probability tables*, then normalize whole rows exactly as rationals; unsupported IDs and zeros are retained. The admitted bridge uses immutable deterministic JSON lexer/predictive-stack sources. Explicit stacks can grow exponentially; preparation is charged and is not the polynomial chart implementation of Theorem 3.

Each exact normalizer is independently checked against the frozen stack posterior. The forest contains 109,026,321,152 valid original-token assignments. Enumerating that count is not attempted or used as a timeout competitor. Independent tiny original-ID enumeration validates the bridge, both masses and TV interval before the public audit.

Two inexpensive competent controls use the same conditioned distributions. If $a \le Q/P \le b$ on P's support and Q is supported there, the sharp range upper bound is $(1-a)(b-1)/(b-a)$, with

the equal-law case handled separately. The Hellinger control uses $\mathrm{TV} \le \sqrt{1 - (\sum_y \sqrt{P(y)Q(y)})^2}$; integer square-root intervals on atom weights and scalar inside inference give a certified rational upper bound. It does not rely on unverified floating-point cancellation.

Format	ε	L	U	Hellinger upper	Core (s)
FP32	1/2	4.894e-09	7.340e-09	8.138e-09	3.862
FP32	1/10	4.894e-09	5.383e-09	8.138e-09	12.282
FP16	1/2	3.723e-05	5.584e-05	2.784e-03	1.109
FP16	1/10	3.723e-05	4.095e-05	2.784e-03	1.935
BF16	1/2	2.061e-04	3.092e-04	4.389e-04	3.440
BF16	1/10	2.061e-04	2.267e-04	4.389e-04	10.526

Table 1: All predefined public numerical audits. Seconds describe the core, not model inference or a competitive speed gate. Rational source rows produce these rounded display values.

For FP16 at $\varepsilon = 0.1$, $U = 4.09523 \times 10^{-5}$, while the Hellinger upper is 67.98 times larger. The table zeros 187,422 original entries by underflow, all kept explicitly. Thus near-tight drift can certify a stricter budget than these controls on this real input. No application budget was selected as a prospective success threshold: this is the complete certificate curve, not a claim of a predeclared semantic win. The controls are cheaper; Table 1 does not imply that our audit should replace them when their bounds already suffice.

6 Related alternatives and exact claim boundaries

Fu's general admissible-gate theorem already permits the disjoint union and componentwise product operations of our forest [1]. Therefore the generic existence of deterministic approximation on an admitted forest is a specialization, rather than our main discovery. The smaller integer core and resulting quantitative bound are the claim of this draft. Specific priority is based on the primary results inspected as of this draft's date, not an exhaustive-literature guarantee.

Domain reduction for smooth decomposable probabilistic circuits requires a common v-tree in its current theorem; determinism is not required [5]. A parse forest can have differing product splits. This distinction is not an encoding-independent separation, and the more general admissible-gate framework remains a competitor. Ordinary exact enumeration is preferable for small valid sets. Certified Hellinger/KL or range bounds are preferable when their cheaper upper guarantees suffice.

The ICML inference reduction [4] also deserves a competent grammar encoding. Normalize each reachable alternative by exact inside masses; activate the children selected by their parents, and give unused nodes a canonical inactive value. Decomposability precludes repeated nonempty nodes in a derivation. Complete choice vectors and valid assignments are bijective. Boolean activation gadgets avoid exponential CPT descriptions for shared nodes. For the reduction's diagonal and topological-prefix queries, minimum local conditional probabilities and clamped already active choices are ordinary weighted inside operations. A visited node's activation is determined by the prefix, so arbitrary future activation evidence is unnecessary. This is a classical randomized alternative, not a newly claimed reduction or a measured implementation. Its failure parameter and ε^{-2} sample dependence differ from the deterministic bound here. Recent linear-time randomized product algorithms are also stronger choices in their setting [6].

EPIC and commitment selectors perform different operations. This manuscript neither evaluates EPIC's complete law nor claims generation superiority over it. Exact token lifting, geometric queries and selected Lean specifications elsewhere in the project do not establish academic novelty by themselves. No formal proof of this complete geometric theorem or Python implementation, independent human review, training improvement or editorial acceptance is claimed.

7 Reproducibility and unresolved obligations

The attempt directory preserves every source version, command and result, including the two input-setup failures and the interrupted slow-query audit. The final numerical run completed all six unchanged cases. Independent checks include complete norm-breakpoint inequalities, exhaustive segment projections, original-ID grammar enumeration and the direct product recurrence; none is treated as evidence of theoretical priority. The packet and frozen helpers have SHA256 provenance. The geometry uses Python integers without model dependencies. Declared resource refusals issue no certificate.

The publication claim needs independent checking of the complete proof and the nearest-prior comparison. Practical generalization beyond this one packet requires relevant new inputs and a prospective protocol. These are not filled by the displayed favorable bound. The present result provides a mathematical precision-cost improvement and a real distribution-audit use, not a promise that every structured decoder becomes faster or more accurate.

References

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